

Alibaba E-commerce database

Chapter - 1

1.1 Introduction

Alibaba.com is one of the world's largest business-to-business (B2B) e-commerce platforms, connecting wholesale buyers with manufacturers and suppliers, primarily based in China and other parts of Asia. Unlike business-to-consumer (B2C) platforms such as Amazon or Flipkart, Alibaba is designed for bulk transactions, allowing retailers, distributors, and entrepreneurs to source products directly from factories at competitive, volume-based pricing.

This report examines JBL-branded Wireless Bluetooth Earbuds as a representative product category on the platform. JBL was chosen because it is a globally recognized audio brand (owned by Harman International), and its earbuds attract heavy buyer interest on Alibaba — both as officially licensed distributor stock and, more commonly, as OEM/"JBL-style" units manufactured by independent factories. This dual reality makes it a particularly useful case study for understanding brand verification and authenticity risk in B2B sourcing, alongside the standard pricing and MOQ dynamics.

1.2 Problem Statement

Alibaba.com is a large-scale B2B wholesale e-commerce platform where thousands of suppliers list products (such as JBL-branded/OEM Bluetooth earbuds) with varying MOQs, prices, and sales volumes. Currently, buyers and small businesses have no structured way to track, compare, and analyse supplier data — such as pricing tiers, minimum order quantities, units sold, and repeat-customer trends — across multiple listings in one place. Manually going through individual product pages to compare suppliers is time-consuming and error-prone, making it difficult to make informed sourcing decisions.

This project aims to design and implement a relational database system that models a wholesale e-commerce platform (using JBL Bluetooth earbuds as a case-study product). The system will store and manage data related to Suppliers, Products, Customers/Buyers, and Orders, allowing users to query and analyse information such

as which supplier offers the best price-to-MOQ ratio, which listings have the highest sales volume, and how customer purchase patterns vary across suppliers — thereby demonstrating how a structured database can simplify sourcing decisions on a B2B wholesale platform.

1.3 Objectives of the Study

- To understand how a wholesale product listing functions on the Alibaba platform.
- To study the pricing structure, MOQ patterns, and supplier landscape for Bluetooth earbuds.
- To analyse how sales volume and customer/buyer numbers typically grow over a sourcing cycle.
- To evaluate the factors that influence a buyer's decision when purchasing wholesale on Alibaba.
- To draw insights on how small businesses can use B2B e-commerce platforms for cost-effective sourcing.

1.4 Motive / Purpose of the Topic

The motive behind studying this topic is to build a practical understanding of how global B2B trade operates in the digital era. Traditional wholesale sourcing required physical trade fairs, agents, and long negotiation cycles. Platforms like Alibaba have digitized this process, giving even small entrepreneurs direct access to manufacturers. Studying a real product category helps illustrate core e-commerce and supply chain concepts — pricing tiers, order volume economics, supplier , applied way rather than as abstract theory

1.5 Stake Holders

1. Suppliers/Manufacturers

Businesses (e.g., Shenzhen/Guangzhou/Dongguan-based trading companies) that list

products, set MOQs and pricing, and full fill bulk orders. They benefit from the database by tracking their own sales performance and visibility.

2. Wholesale Buyers/Customers

Retailers, distributors, or entrepreneurs sourcing products in bulk. They use the system to compare suppliers, prices, and MOQs to make informed purchasing decisions.

3. Alibaba Platform (Marketplace Administrator)

Acts as the intermediary that hosts suppliers and buyers, verifies suppliers (Gold Supplier, Trade Assurance badges), and maintains transaction/order records.

4. Database Administrator/Developer

Responsible for designing, maintaining, and optimizing the database schema — ensures data integrity between Suppliers, Products, Customers, and Orders tables.

5. Academic Evaluator/Guide (for your skill project)

Your course instructor/HOD who assesses the project's design, implementation, and query outcomes as part of the 3rd sem curriculum.

6. End Analysts/Decision-makers

Anyone (student, business analyst, or small business owner) using the query outputs to identify best-performing suppliers, popular products, or purchasing trends.

1.6 Business Requirements

- The system must maintain records of all suppliers, including business name, location, MOQ policy, certification status (e.g., Gold Supplier, Trade Assurance), and contact details.
- The system must maintain records of all products listed by suppliers, including product name, category, price range, key features, and listing type (OEM/licensed).
- The system must track customers/buyers, including business name, contact information, and purchase history.
- The system must record orders, capturing which customer purchased which product from which supplier, order quantity, order date, and order value.
- The system must support tracking of units sold per listing to identify high-performing vs. low-performing products.

- The system must allow comparison of suppliers based on price, MOQ, and sales volume to help buyers make sourcing decisions.
- The system must support identification of repeat customers and their purchase frequency.
- The system must ensure data integrity through proper primary/foreign key relationships between Suppliers, Products, Customers, and Orders.
- The system must allow querying and reporting, such as top-selling products, best-rated suppliers, and monthly/periodic sales trends.
- The system must be scalable, allowing new suppliers, products, and customers to be added without restructuring the database.
- The system must support data security and access control, ensuring only authorized users (e.g., admin) can modify supplier or pricing data.
- The system should provide a foundation for future analytics/dashboard integration (e.g., visualizing sales trends by supplier or product category).

1.7 Functional Requirements

Supplier Management

- The system shall allow adding, updating, and deleting supplier records (name, location, MOQ policy, certification, contact info).
- The system shall allow searching suppliers by location, certification status, or product category.

Product Management

- The system shall allow adding, updating, and deleting product listings (name, category, price, features, listing type).
- The system shall allow linking each product to one or more suppliers.
- The system shall allow filtering products by price range, MOQ, or category.

Customer Management

- The system shall allow registering new customers/buyers with their contact and business details.

- The system shall maintain a purchase history for each customer.
- The system shall identify repeat customers based on order frequency.

Order Management

- The system shall allow recording new orders, capturing customer, product, supplier, quantity, price, and date.
- The system shall calculate total order value automatically (quantity × unit price).
- The system shall update the "units sold" counter for a product after each successful order.
- The system shall allow order status tracking (e.g., Pending, Confirmed, Shipped, Delivered).

Search & Query Functions

- The system shall allow querying top-selling products across suppliers.
- The system shall allow comparing suppliers by price, MOQ, and total units sold.
- The system shall allow generating a list of repeat buyers for a given product/supplier.
- The system shall allow generating monthly/periodic sales summaries.

Reporting

- The system shall generate supplier performance reports (total orders, total revenue, certification status).
- The system shall generate product performance reports (total units sold, revenue generated).
- The system shall generate customer purchase reports (frequency, total spend).

Validation & Data Integrity

- The system shall enforce primary key/foreign key constraints between Suppliers, Products, Customers, and Orders tables.
- The system shall prevent duplicate supplier/product/customer entries using unique constraints.

- The system shall validate that order quantity does not fall below a product's MOQ.

Access Control

- The system shall restrict supplier and pricing data modification to admin users only.
- The system shall allow read-only access to buyers/customers for browsing product and supplier data.

Scalability

- The system shall support the addition of new suppliers, products, and customers without requiring schema changes.

1.8 Non functionality

Performance

- The system shall retrieve query results (e.g., top-selling products, supplier comparisons) within 2–3 seconds for typical data volumes.
- The database shall efficiently handle at least 10,000+ product and order records without significant slowdown.

Scalability

- The system shall be designed to accommodate growth in suppliers, products, customers, and orders without requiring major schema redesign.

Reliability

- The system shall ensure accurate and consistent data (e.g., units sold counters, order totals) even under concurrent updates.
- The system shall maintain referential integrity at all times (no orphan records in Orders, Products, etc.).

Availability

- The database shall be available for querying and reporting at all times during platform operation (target: 99% uptime in a production scenario).

Security

- The system shall restrict write/update access to sensitive data (pricing, supplier details) to authorized/admin users only.
- The system shall store any sensitive information (e.g., contact details) securely, avoiding plain-text exposure where possible.

Usability

- The database structure and any accompanying interface/queries shall be simple enough for a non-technical buyer or evaluator to understand key reports.
- Table and column names shall be self-descriptive (e.g., `supplier_name`, `units_sold`) for ease of maintenance.

1.9 Conclusion

This skill project demonstrates how a structured relational database can streamline sourcing and sales analysis on a B2B wholesale e-commerce platform like Alibaba.com. Using JBL Bluetooth earbuds as a case study, the project identifies key entities — Suppliers, Products, Customers, and Orders — and defines the business, functional, and non-functional requirements needed to manage them effectively.

By organizing scattered supplier and product information into a normalized database, buyers can easily compare pricing, MOQ, and sales performance across suppliers, while suppliers and platform administrators gain a reliable system for tracking orders and customer behavior. The project also highlights a real-world consideration in B2B sourcing — brand authenticity verification — showing that database design must account for practical business nuances, not just technical structure.